LAB ASSIGNMENT-5

Aim- Regularization Techniques: Ridge and Lasso — Influence on Overfitting and Multicollinearity

1. Problem Statement

Apply regularization techniques such as Ridge (L2) and Lasso (L1) regression to a suitable multivariate open-access dataset, and observe their influence on model overfitting and on multicollinearity among the predictors.

2. Objectives

- To detect and quantify multicollinearity in a multivariate dataset using the correlation matrix, Variance Inflation Factor (VIF) and the condition number of the design matrix.
- To implement and compare Ordinary Least Squares, Ridge, Lasso and Elastic Net regression using Python and scikit-learn.
- To observe how the regularization parameter α shrinks the coefficient estimates and to plot the coefficient paths of Ridge and Lasso.

3. Theory

Multicollinearity

Multicollinearity is the presence of near-linear dependence among the predictor columns of X. It does not necessarily damage predictive accuracy, but it makes the individual coefficients unstable and uninterpretable: the data cannot separate the contribution of one correlated predictor from another. Three standard diagnostics are used.

Consequences of multicollinearity: inflated standard errors, coefficients of implausible magnitude or with the wrong sign, coefficients that swing wildly when a few observations are added or removed, and predictors that appear statistically insignificant even though the model as a whole fits well.

Ridge Regression (L2)

Ridge Regression is a regularization technique used in linear regression to reduce overfitting and handle multicollinearity among input features. It adds a penalty based on the square of the regression coefficients to the ordinary least-squares error. The objective is to minimize:

Cost = Sum of Squared Errors + $\lambda \times$ Sum of $(\text{coefficients})^2$

Here, λ (lambda) controls the strength of regularization. A larger λ produces stronger regularization and shrinks the coefficients closer to zero.

Unlike Lasso Regression, Ridge does not usually make coefficients exactly zero**.** Therefore, it keeps all features in the model but reduces the influence of less important features. For example, if a model has 10 features, Ridge may reduce the coefficients of some features from large values to smaller values, but generally keeps all 10 features. Thus, Ridge is useful when most features contribute to the prediction and when correlated features are present.

Lasso Regression (L1)

Lasso Regression (Least Absolute Shrinkage and Selection Operator) is a regularization technique used in linear regression to reduce overfitting and improve model performance. It adds a penalty term based on the absolute values of the regression coefficients to the ordinary least-squares error. The objective function is:

Cost = Sum of Squared Errors + λ × Sum of |coefficients|

Here, λ (lambda) controls the strength of regularization. When λ is increased, the coefficients are reduced toward zero.

An important feature of Lasso Regression is feature selection. Unlike Ridge Regression, Lasso can make some coefficients exactly zero, effectively removing less important features from the model. Therefore, Lasso is particularly useful when a dataset contains many features and we want to identify the most important ones. For example, if a model has 10 input features and Lasso reduces the coefficients of 4 features to zero, the model effectively uses only the remaining 6 features.

Conclusion

Ridge and Lasso regularization were successfully implemented on the open-access Wine Quality (red) dataset, and their influence on overfitting and on multicollinearity was studied experimentally in both a regression and a classification setting.

Viva Questions

1. What is Lasso Regression, and why is it used?
2. What is Ridge Regression, and why is it used?
3. What is the main difference between Lasso and Ridge Regression?
4. What type of penalty is used in Lasso Regression?
5. What type of penalty is used in Ridge Regression?
6. Why can Lasso Regression perform feature selection?
7. What is the role of the λ (lambda) parameter in Lasso and Ridge Regression?